WORKED SOLUTION

STEP II 1988, Question 9

Pure • Subgroups and Unions

STEP II 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q9

This question has a full guided treatment in *Beyond STEP*, Section 6.6, *Unions of Subgroups*: https://beyondstep.keithmcnulty.org/ch06/subgroup-unions.html

The key idea

Argue by contradiction. If neither subgroup is the whole group, pick an element that only G_2 contains and an element that only G_1 contains, then ask where their product lives. Closure under multiplication and inverses gives a contradiction whichever subgroup we put it in. With three subgroups the product has somewhere else to go. The natural group to try is one in which every element is its own inverse.

Solution

Two subgroups

Suppose that every element of G lies in G_1 or in G_2, and suppose, for a contradiction, that $G_1 \neq G$ and $G_2 \neq G$.

Since $G_1 \neq G$ and $G_1 \subseteq G$, there is an element $a \in G$ with $a \notin G_1$. Every element lies in G_1 or G_2, so $a \in G_2$.

In the same way, since $G_2 \neq G$ there is an element $b \in G$ with $b \notin G_2$, and then $b \in G_1$.

So we have

$$a \in G_2, \quad a \notin G_1, \qquad\qquad b \in G_1, \quad b \notin G_2.$$

Now look at the product $ab \in G$. By hypothesis it lies in G_1 or in G_2.

- If $ab \in G_1$: since $b \in G_1$ and G_1 is a subgroup, $b^{-1} \in G_1$, and so by closure

$$a = (ab)\, b^{-1} \in G_1,$$

  which contradicts $a \notin G_1$.

- If $ab \in G_2$: since $a \in G_2$, also $a^{-1} \in G_2$, and so
$$b = a^{-1}(ab) \in G_2,$$
which contradicts $b \notin G_2$.

Both cases are impossible, so our assumption was false: $G_1 = G$ or $G_2 = G$. ■

Ask yourself Where did we use that G is finite?

We didn't. The argument uses only the subgroup properties (closure and inverses) and works for any group, finite or infinite. Equivalently: a group is never the union of two proper subgroups.

Three subgroups

Take H to be the Klein four-group

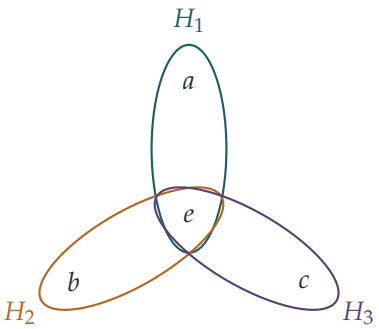
$$H = \{e,\ a,\ b,\ c\}, \qquad a^2 = b^2 = c^2 = e, \quad ab = ba = c,\ bc = cb = a,\ ca = ac = b.$$

Concretely, H is $\{(0,0),(1,0),(0,1),(1,1)\}$ under addition modulo 2 in each coordinate, or the symmetry group of a non-square rectangle (identity, two reflections and the half-turn).

Every non-identity element has order 2, so each of

$$H_1 = \{e,a\}, \qquad H_2 = \{e,b\}, \qquad H_3 = \{e,c\}$$

is a subgroup: it contains e, is closed under multiplication (the only non-trivial product is $a \cdot a = e$, and similarly for b and c), and each element is its own inverse. Each has 2 elements, so none of them is H. But between them they contain e, a, b and c, which is every element of H.



So three proper subgroups can cover a group. Another example is the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, which is the union of the three cyclic subgroups $\{\pm 1, \pm i\}$, $\{\pm 1, \pm j\}$ and $\{\pm 1, \pm k\}$, each of order 4.

Notice why the first argument breaks down here: with $a \notin H_2$ and $b \notin H_1$, the product $ab = c$ is free to escape into the third subgroup H_3.

Checks

- A brute-force search over every subgroup of the cyclic groups of order 1 to 12, of S_3, S_4, the dihedral group of order 8, Q_8, $\mathbb{Z}_2^2$ and $\mathbb{Z}_2^3$ (1278 ordered pairs of proper subgroups) found no pair whose union is the whole group.
- The same search confirms that $\{e, a\} \cup \{e, b\} \cup \{e, ab\}$ is all of the Klein four-group, and Q_8 is the union of $\langle i \rangle$, $\langle j \rangle$, $\langle k \rangle$. Interestingly, none of the cyclic groups tested, nor S_3 or S_4, is the union of three proper subgroups.
- A quick sanity check on the first part: in a cyclic group, a generator lies in no proper subgroup at all, so a cyclic group can't even be covered by any number of proper subgroups.

Common slips

- Arguing by counting: two proper subgroups each have at most $\frac{1}{2}|G|$ elements, but they share e, so their union has at most $|G| - 1$ elements. This counting proof is correct (by Lagrange's theorem) and short, but it does need finiteness and Lagrange; make sure you state both.
- Picking $a \notin G_1$ and $b \notin G_2$ but forgetting to use that $a \in G_2$ and $b \in G_1$. Without those two facts the contradiction doesn't follow.
- Offering three subgroups that are not all proper, or not subgroups at all (for example $\{e, a, b\}$, which is not closed since $ab = c$).

Beyond STEP

Which groups are the union of three proper subgroups? A theorem of Scorza (1926) gives the answer: exactly those with a quotient isomorphic to the Klein four-group, which is why the Klein group is the basic example. The same "no union of two" fact appears in linear algebra, where a vector space over an infinite field is never a finite union of proper subspaces. For a first course in groups, try Allenby, *Rings, Fields and Groups*.

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